

Recognizing and Translating Between Portuguese Varieties with Fine-tuned Language Models

PIC2 - Master in Computer Science and Engineering
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Abstract Large language models trained on Portuguese text are strongly shaped by the predominance of Brazilian Portuguese (BP) on training corpora collected from the Web, which can reduce their effectiveness for European Portuguese (EP). Effective methods for filtering and/or translating between both Portuguese variants are thus in need. This work aims to explore how decoder-only pre-trained models can be adapted to distinguish between the BP and EP variants, and to rewrite text from one variant to the other. We will construct a unified corpus by combining OPUS OpenSubtitles data with other datasets prepared in previous studies, and we will apply a tailored preprocessing pipeline that includes normalization, sentence merging, boilerplate removal, deduplication, sentence-length filtering, and alignment filtering. Using this corpus, we plan to adapt multilingual language models to support both the language variety identification and translation tasks. We have already conducted an initial experiment with a model of the Qwen family. This work aims to provide a reproducible methodology and a consolidated resource for developing large language models that handle Portuguese varieties more reliably.

Keywords — Language Variety Identification; Large Language Models

*I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa (<https://nape.tecnico.ulisboa.pt/en/apoio-ao-estudante/documentos-importantes/regulamentos-da-universidade-de-lisboa/>).

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1 Introduction

Language modeling seeks to estimate the generative likelihood of word sequences to predict future or missing tokens. At large parameter scales, pre-trained Large Language Models (LLMs) often exhibit strong and sometimes emergent abilities across diverse text analysis tasks. However, for languages with multiple standardized varieties, models trained in one variety can fail in another. Among the possible downstream applications, discriminating or translating between language varieties is particularly important in Natural Language Processing (NLP). LLMs can be applied to these problems, aiming to improve the experience of users who rely on different varieties of the same language.

In real-world settings, populations evolve and migrate, and usage patterns mix across regions and domains, which complicates the discrimination and also the reliance on strictly variety-specific models. However, if dominant varieties drive training corpora and model behaviour, communities using minority varieties are less well served by LLM-based systems (Sousa et al., 2025b).

In the specific case of Portuguese, distinguishing European Portuguese (EP) from Brazilian Portuguese (BP) remains challenging due to nuanced differences across orthography and grammar, as well as divergent domain-specific vocabulary that is only used in specific contexts. This problem is particularly important given that around 200 million people speak Portuguese and it is the official language of eight nations, each with its own variety. However, roughly 70% of Portuguese speakers are Brazilian, so resources labelled as “Portuguese” are typically dominated by the Brazilian variety, which risks biasing models toward that variety (Sousa et al., 2025b). Beyond data imbalance at the point of collection, any classifier-based filtering used during LLM pre-training can inadvertently remove dialectal and sociolectal text, reducing diversity and further biasing any filtered corpora. Although public machine translation services now expose EP and BP options, support for direct EP-BP translation remains limited.

This work envisions three main contributions:

- A consolidated EP-BP corpus: we plan to construct unified resources by combining OPUS Open-Subtitles (Tiedemann, 2009), FRMT (Riley et al., 2023), the Gold Collection (Sanches et al., 2024), and PtBrVarId (Sousa et al., 2025b), together with a documented preprocessing and filtering pipeline.
- A reusable variety-aware model: we will adapt decoder-only language models to perform EP-BP rewriting and EP/BP identification within a unified instruction-following formulation, producing a model that can be reused directly and can also support improving larger models via additional fine-tuning or test-time scoring and re-ranking.
- Validation and analysis: we aim to validate both the corpus and the resulting models through quantitative evaluation and targeted qualitative error analysis, with emphasis on robustness under domain shift and compliance with explicit EP/BP preferences.

Concretely, we have already produced an initial aggregated version of the dataset and report its corpus statistics in this document. Construction relies on a dedicated preprocessing pipeline that includes normalization, deduplication, and other cleaning steps across sources. For OPUS in particular, the pipeline incorporates subtitle-specific quality controls: a sentence-length outlier analysis identified 161,854 pairs to remove, and a SimAlign-based alignment filter (Sabet et al., 2021) was applied to further improve pair quality.

From a modeling perspective, we plan to fine-tune decoder-only language models using parameter-efficient methods. Initial experiments have been conducted with a Qwen3 model of 0.6 billion parameters, primarily to validate the feasibility of the training formulation and pipeline. Given the available compute budget and the strong empirical performance often reported for Parameter-Efficient Fine-Tuning (PEFT) (Houlsby et al., 2019), we plan to rely primarily on Low-Rank Adaptation (LoRA) adapters (Hu et al., 2021). All code is available online.¹

During the second semester, the work will focus on consolidating and strengthening both the data and modeling components. On the data side, we plan to refine the corpus-construction pipeline by revisiting

¹<https://github.com/laiarodrigo/Thesis>

OPUS preprocessing and the sentence-length and alignment-based filters, and by updating descriptive statistics once the final filtering strategy is fixed. On the modeling side, we plan to improve the training setup and broaden the set of evaluated models, including systematic comparisons between PEFT and full fine-tuning under matched budgets. Finally, we plan to explore preference-based adaptation by using a variety-aware model as a scorer for re-ranking and as a reward signal in Reinforcement Learning (RL) style optimization, with the objective of producing systems that follow EP or BP preferences reliably when explicitly requested.

The remainder of this document is organized as follows. Section 2 reviews key background concepts, including contrasts between European and Brazilian Portuguese, language-model architectures, Transformers, and fine-tuning techniques. Section 3 offers a comprehensive literature review on the state-of-the-art relating to EP–BP classification and translation, as well as for other language pairs. Section 4 details corpus construction and preprocessing, including the OPUS pipeline, the alignment-based quality filter with SimAlign, the PtBrVarId pipeline, and reports corpus statistics and quality checks for all the datasets that were aggregated. Section 5 formulates the hypothesis that this work seeks to validate, showcases our initial experiments with corpus construction and our plans to refine it, the modeling strategy we plan to use in the next semester, and covers preliminary results. Finally, Section 6 presents a summary of the report and outlines the calendarization of the work for the next semester.

2 Fundamental Concepts

This section outlines the theoretical and technical foundations that underpin our study and establishes the terminology used throughout. Section 2.1 summarizes the main linguistic distinctions between European and Brazilian Portuguese that are relevant for computational modeling, highlighting recurring orthographic, morphosyntactic, and lexical phenomena that motivate variety-aware adaptation. Section 2.2 then introduces language models and traces the progression from statistical approaches to neural architectures and, ultimately, the Transformer paradigm that underlies contemporary LLMs. Section 2.3 provides a focused view of Transformer mechanics. Building on this background, Section 2.4 reviews the principal language-model families and discusses their suitability for classification and rewriting tasks. Section 2.5 presents standard fine-tuning strategies, with particular emphasis on parameter-efficient methods. Finally, Section 2.6 introduces reinforcement learning and preference-based fine-tuning for language models.

2.1 Differences Between European Portuguese and Brazilian Portuguese

Public Portuguese text available online is dominated by BP, and thus base models often mirror BP spelling, morphology, and syntactic preferences. For tasks that require distinguishing varieties and rewriting between them, targeted adaptation is necessary. The fine-tuning corpus must therefore expose contrasts at multiple linguistic levels (orthography, morphosyntax, and lexicon), including contrasts that are robustly observable in text and frequently exploited in computational work (Costa-jussà et al., 2018; Marujo et al., 2011).

Prior literature describes these differences both through qualitative grammatical analysis and through resources designed for automatic rewriting. For instance, Barreiro and Mota (2018) extract and manually align EP-BP paraphrastic correspondences from parallel translations, showing that variety contrasts often surface as systematic alternatives between expressions and constructions. Their aligned examples highlight recurring differences in progressive aspect, clitic placement, determiner realization, subject realization, forms of address, and lexical choice (e.g., *comboio* versus *trem* or *desporto* versus *esporte*) (Barreiro and Mota, 2018). Cyrino (2001) discusses contrasts involving null objects and verb phrase ellipsis, relating them to changes in the verbal structure of BP sentences.

The differences below illustrate several recurrent, corpus-visible phenomena that are particularly relevant for modeling variety identification and controlled rewriting:

- **Orthography and morphology:** Differences in spelling persist, although several contrasts were reduced by the Portuguese Orthographic Agreement. In particular, older European Portuguese

texts may still retain muted consonants (e.g., *projecto*, *acto*, *baptismo*), whereas contemporary standardized spelling typically drops them (e.g., *projeto*, *ato*, *batismo*). Such forms can still appear in legacy corpora and archival material, but are less reliable markers in post-agreement data.

- **Progressive and aspectual periphrases:** European Portuguese typically favors the progressive with *estar a* + infinitive (e.g., *Ele estava a correr*), whereas Brazilian Portuguese strongly prefers *estar* + gerund (e.g., *Ele estava correndo*) (Barreiro and Mota, 2018).
- **Clitic placement and word order:** European Portuguese commonly attaches clitics postverbally with a hyphen in affirmative contexts (e.g., *Ele viu-me na rua*), while Brazilian Portuguese more often uses preverbal placement without hyphenation (e.g., *Ele me viu na rua*) (Barreiro and Mota, 2018; Costa-jussà et al., 2018).
- **Subject pronoun realization (pronoun drop):** Both varieties allow null subjects, but BP more frequently includes subject pronouns in contexts where EP more readily omits them. For example, EP commonly allows a null subject (e.g., *Fui ao supermercado ontem*), whereas BP more often uses an overt pronoun (e.g., *Eu fui ao supermercado ontem*) (Barreiro and Mota, 2018).
- **Pronoun preferences:** Brazilian Portuguese frequently uses *você* as the default 2nd-person address form, while European Portuguese more often uses *tu* in informal contexts (or omits the overt subject), and reserves *você* for more restricted registers. In practice, this yields frequent rewrite pairs such as EP 2nd-person verb morphology vs. BP 3rd-person morphology with *você* (Barreiro and Mota, 2018; Costa-jussà et al., 2018).
- **Determiners with possessives and zero determiners:** In Brazilian Portuguese, definite articles before possessives are often omitted (e.g., *Vendi meu carro*), whereas European Portuguese typically requires them (e.g., *Vendi o meu carro*). More generally, BP omits more determiners in contexts where EP prefers an overt article, a pattern explicitly discussed in paraphrase alignment work (Barreiro and Mota, 2018; Cyrino, 2001).
- **Lexical and multiword preferences:** The largest and most reliable distinctions in written corpora often come from lexical choice and multiword expressions (e.g., *comboio* versus *trem*, *fato* versus *terno*), as well as compositional alternatives (e.g., *toda a gente* versus *todo o mundo*) (Barreiro and Mota, 2018; Costa-jussà et al., 2018).

Complementing qualitative descriptions, Alves (2025) applied information-theoretic measures to the FRMT dataset to characterize differences between EP and BP. The author showed that the largest divergence arises at the lexical level, due to terms that are strongly associated with one variety or the other, but also uncovered systematic grammatical preferences, such as differences in the use of subject pronouns, progressive constructions, and clitic position. An analysis of Part-Of-Speech (POS) distributions, dependency relations, and POS trigrams further highlighted distinctive constructions, while word-order entropy indicated that most syntactic patterns are shared but certain preferences remain. More specifically, word-order entropy measures how predictable the linear order of syntactic units is: lower entropy indicates more fixed word order, while higher entropy suggests greater flexibility.

These contrasts motivate the need for models that can both recognize variety-specific cues and generate controlled rewrites across varieties. The next subsection therefore provides a brief overview of language models, introducing the main modeling paradigm and the architectural developments that led to contemporary Transformer-based systems.

2.2 Language Models

A language model assigns probabilities to a sequence of tokens and uses the surrounding context to predict what will follow (Zhao et al., 2025). In practice, language models are trained so that continuations observed in the data become more likely, for instance by minimizing the average negative log-likelihood

(cross-entropy) over large text corpora. At inference time, the very same predictor can generate text one token at a time, or score how well a given sequence fits the model.

The path to today’s systems moved through several phases. Early statistical models relied on short, fixed windows of context, leading to n -gram models that were fast but limited to local patterns (Rosenfeld, 2000; Jelinek, 1997). Neural language models (Bengio et al., 2003) replaced counts with learned representations, first with feed-forward networks over fixed contexts and then with recurrent networks that could, in principle, track longer dependencies (Mikolov et al., 2010). A major shift occurred with Transformers (Vaswani et al., 2023), which use self-attention to make long-range context directly accessible. Pre-training on large unlabeled corpora followed by task-specific adaptation has since become a standard paradigm for deploying language models across a broad range of NLP tasks.

BERT (Devlin et al., 2019) crystallized Transformer-based encoder models for NLP, where bidirectional masked-token pre-training then transfers across classification and retrieval tasks with minimal task-specific parameter changes.

As parameters, data diversity, and context windows grew, decoder-only Transformers began to generalize widely from prompts and/or through modest adaptation. At large scales, models started to exhibit abilities such as robust in-context learning and multi-step reasoning (Zhao et al., 2025).

Since the models considered in this work are Transformer-based, the next subsection reviews the self-attention mechanism and the encoder/decoder variants that underpin Transformer architectures.

2.3 Transformers: Self-Attention, Encoders and Decoders

To explain in more detail what a Transformer is, we start from the self-attention operation. Self-attention lets each token in a sequence mix information from all other tokens by computing attention weights that reflect learned similarities. Each token is mapped to a continuous embedding. Since self-attention is permutation-invariant, token order is injected through positional encodings. In the original Transformer these are fixed sinusoidal vectors added to token embeddings, yielding input representations $X \in \mathbb{R}^{n \times d_{\text{model}}}$. The sequence length n is limited by the model’s context window, i.e., the maximum number of tokens it can attend to in one forward pass.

Given token representations X , which include positional information, we obtain queries, keys, and values via linear projections $Q = XE_Q$, $K = XE_K$, and $V = XE_V$. The scaled dot-product attention used throughout the architecture is given by

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

so each token representation is replaced by a similarity-weighted average of the others. Multi-head attention repeats this computation with different learned projections and concatenates the results so that the model can capture multiple interaction patterns in parallel, making transformers easy to adapt in a big variety of tasks (Vaswani et al., 2023).

Despite the existence of different architectures using self-attention, the original Transformer consisted of an encoder-decoder architecture and aimed to perform sequence-to-sequence tasks. An encoder is a stack of Transformer blocks, each with a self-attention sublayer followed by a position-wise feed-forward network, wrapped with residual connections and layer normalization for stable training. The feed-forward network is position-wise in the sense that it applies the same multilayer perceptron to each token representation independently, refining features at each position without exchanging information across positions. Since encoder self-attention is bidirectional, every position can attend to every other position in the input, yielding contextual representations that are useful for understanding and for conditioning downstream generation (Vaswani et al., 2023).

A decoder is also a stack of Transformer blocks, but it adds two ingredients required for output generation. First, masked self-attention applies a causal mask, a triangular masking pattern that sets attention to future positions to zero, so the model can generate left to right. Second, a cross-attention sublayer attends to the encoder’s outputs, allowing the decoder to condition on the source sequence while producing the target sequence.

This architectural template gives rise to distinct model families, which differ in training objectives and in how naturally they support classification and text generation. The next subsection reviews these families and motivates the model selection for the initial experiment performed in this work.

2.4 Language Model Families Amenable to Fine-tuning

Modern approaches implement the same core principle mentioned in the previous sections, which consists in predicting tokens under context using Transformer-based architectures. In practice, models typically fall into three families: encoder-only, encoder-decoder, and decoder-only.

Encoder-only models, such as BERT (Devlin et al., 2019), are bidirectional encoders trained to reconstruct masked tokens or spans. After self-supervised training, these models can be fine-tuned with task-specific heads for classification, retrieval, or span prediction. They produce high-quality sentence and document representations but are not generators by default, so using them for EP-BP rewriting (i.e., a generative task) requires the attachment of a separate decoder or switching architectures, which complicates training and deployment (Raffel et al., 2023).

Encoder-decoder models encode an input and decode an output autoregressively, conditioned on the representation. This setup is the classic choice for machine translation, since the decoder can attend to the encoder’s representation of the source.

Decoder-only models use a single Transformer stack with a causal mask. They can generate text by conditioning on a prompt and generating the continuation, and classify either by generating a label or by scoring each label with the model’s conditional likelihood. As generation is their native operation, these models can naturally handle free-form production, stepwise reasoning, and flexible reformulation tasks such as summarization, style transfer, and variety rewriting.

All three families operate over tokenized text with a fixed context window, and their capabilities expand with better data, more parameters, and longer contexts. In our setting, which involves distinguishing European from Brazilian Portuguese and rewriting between the two variants, the choice of architecture affects how naturally a single model can cover both tasks. In practice, a decoder-only architecture gives one unified interface for classification (via scoring) and rewriting (via generation) and works smoothly with PEFT techniques.

Accordingly, we focus on decoder-only families whose weights and tooling make them practical to adapt under tight compute. Some of the most widely used model families are LLaMA, Qwen, Gemma, GPT, and Mistral, with models available at different sizes. Recent research efforts at the European level have also led to small models such as MiniLingua (Aksenova et al., 2025), which can be particularly interesting to our work. MiniLingua is an open-source multilingual model of approximately one billion parameters, trained from scratch for 13 European languages and released together with its weights, tokenizer, and training artifacts. The authors report competitive results on multilingual and instruction-following evaluations, including improvements over comparators trained with larger budgets.

Among the well established model families, Qwen offers strong multilingual coverage and provides small models, at scales such as 0.6B, 1.5B, and 3B parameters, making them attractive bases when memory is scarce (Qwen et al., 2025). We therefore considered a Qwen model in our initial experiments.

In particular, we use Qwen3, a decoder-only family that follows the standard Transformer decoder blueprint but incorporates several implementation choices that have become common in modern LLMs. To make the subsequent fine-tuning setup and computational trade-offs clear, we briefly summarize the architectural elements of Qwen3 that differ from the original Transformer formulation.

First, instead of fixed sinusoidal positional encodings, Qwen3 uses Rotary Positional Embeddings (RoPE), which encode positions by applying relative rotations to query and key vectors (Su et al., 2023). This way of creating embeddings lets the model represent relative distances directly in attention scores and extrapolate to long contexts.

Second, Qwen3 replaces LayerNorm with RMSNorm (Jiang et al., 2023), which instead of centering activations and normalizing by their variance as in LayerNorm, normalizes only the root-mean-square (RMS) of each hidden vector. RMSNorm has a simpler parameterization and has been observed to improve stability in deep, large-scale models.

Finally, Qwen3 adopts grouped-query attention (Ainslie et al., 2023) instead of assigning separate key and value projections to every attention head. Multiple query heads share a smaller set of key-value heads, reducing memory while still allowing diverse attention patterns. In the 0.6B model we plan to use, this yields an asymmetric configuration with more query heads than key-value heads per layer.

2.5 Efficient Language Model Fine-tuning

Given the model families discussed above, the next step is to consider how a general pre-trained model can be adapted to a specific task. Fine-tuning is the standard mechanism for this adaptation, enabling the model to better match the desired behavior using comparatively small amounts of task-relevant data.

Starting from a pre-trained language model, we can continue training on smaller, task-specific data so the outputs better match the intended goal. In supervised fine-tuning (SFT), we use labeled examples and optimize the task loss. When labeled data are scarce, unsupervised adaptation continues the language-modeling objective on in-domain text or task-adaptive pretraining, which reliably improves downstream performance (Raffel et al., 2023).

A central decision when fine-tuning a model is the number of parameters to update. Full fine-tuning updates all model weights and can deliver the strongest specialization, particularly under large domain shifts, at the cost of high compute, memory, and a greater risk of overfitting when supervision is narrow. Parameter-Efficient Fine-Tuning (PEFT) keeps the pretrained backbone frozen and learns a small number of additional parameters. Adapter-based methods can be trained to perform multiple tasks inserting a small number of task specific parameters, while keeping the underlying pretrained language model fixed, achieving accuracy close to full fine-tuning while adding a modest number of task-specific parameters (Houlsby et al., 2019; Pfeiffer et al., 2021). LoRA instead learns low-rank updates to selected weight matrices while leaving the original weights untouched. These updates can be merged back into the base weights at deployment, so inference cost remains essentially unchanged. Quantized LoRA further reduces memory by holding the frozen backbone in low precision during training while keeping the learned adapters in higher precision, enabling efficient fine-tuning of very large models on limited hardware.

While the methods above describe how to adapt a model, the next subsection focuses on the used objectives when fine-tuning for alignment with preferences and human feedback (Ouyang et al., 2022).

2.6 Reinforcement Learning and Preference-Based Fine-Tuning

In preference-based fine-tuning of language models, it is convenient to view text generation as a contextual bandit problem (Sutton and Barto, 2018; Lattimore and Szepesvári, 2020) rather than a full Markov Decision Process (MDP). Classical multi-armed bandits model a situation where an agent repeatedly chooses between several actions with unknown reward distributions and, after each choice, observes only the reward for the action it took. There is no notion of state transitions or long-term planning, each round is independent, and the goal is simply to learn which actions tend to give higher rewards.

Contextual bandits extend the multi-armed bandit setting by conditioning each decision on side information. At each round, the environment draws a context X from a distribution $\mathcal{D}$, the agent selects an action $a \sim \pi(\cdot | X)$, and observes a scalar reward $r(X, a) \in \mathbb{R}$. The objective is to learn a policy π that maximizes expected reward,

$$J(\pi) = \mathbb{E}_{X \sim \mathcal{D}} \left[\mathbb{E}_{a \sim \pi(\cdot | X)} [r(X, a)] \right].$$

In our case, the context X corresponds to an instruction together with a source sentence, and the action is a generated output text. Crucially, in a contextual bandit there are no state transitions driven by intermediate actions: the context X does not evolve as a consequence of the chosen output.

In LLM post-training, we write the policy as a conditional distribution over complete outputs $Y = (Y_1, \dots, Y_{|Y|})$ given the context X . A decoder-only language model defines an autoregressive policy

$$\pi_{\theta}(Y | X) = \prod_{t=1}^{|Y|} \pi_{\theta}(Y_t | X, Y_{<t}).$$

After sampling an output $Y \sim \pi_\theta(\cdot | X)$, we evaluate it with a scoring function (reward model, verifier, or preference scorer) that returns a single outcome-level reward $r(X, Y)$. The contextual-bandit objective for policy optimization then becomes

$$J(\pi_\theta) = \mathbb{E}_{X \sim \mathcal{D}} \left[\mathbb{E}_{Y \sim \pi_\theta(\cdot | X)} [r(X, Y)] \right].$$

In this setting, reinforcement learning, directly optimizes the expected-reward objective $J(\pi_\theta)$ by updating θ . Intuitively, the policy is trained to make high-reward outputs more likely under each context X .

Once this view is adopted, standard policy-gradient methods apply. Proximal Policy Optimization (PPO) (Schulman et al., 2017) is a widely used on-policy method that updates the policy using samples from an older snapshot $\pi_{\theta_{\text{old}}}$. Defining the likelihood ratio

$$\rho_t(\theta) = \frac{\pi_\theta(Y_t | X, Y_{<t})}{\pi_{\theta_{\text{old}}}(Y_t | X, Y_{<t})},$$

PPO maximizes the clipped surrogate objective

$$J_{\text{PPO}}(\theta) = \mathbb{E} \left[\frac{1}{|Y|} \sum_{t=1}^{|Y|} \min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right],$$

where $\hat{A}_t$ is an advantage estimate that indicates how much better the sampled output is than a baseline under the same context. Here, $\text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)$ truncates the likelihood ratio so that each update cannot change the probability of the sampled tokens too aggressively. This stabilizes training by preventing very large policy updates from a small number of high-advantage samples.

In many LLM post-training setups, PPO is paired with a regularization term that keeps the learned policy close to a fixed reference model π_{ref} , typically through a KL penalty,

$$-\beta D_{\text{KL}}(\pi_\theta(\cdot | X) \parallel \pi_{\text{ref}}(\cdot | X)),$$

which discourages the policy from drifting too far from the supervised fine-tuned behaviour. The KL term measures how far the updated policy deviates from a fixed reference policy (typically the supervised fine-tuned model).

Alternatively, Direct Preference Optimization (DPO) (Rafailov et al., 2024) takes a different approach. DPO operates on logged preference pairs collected in advance. For each input, it assumes access to a preferred output and a not preferred one and adjusts the policy so that, relative to a reference model, it assigns higher probability to the preferred output and lower probability to the other. This is implemented through a simple logistic loss on differences of log-probabilities. Conceptually, DPO still has a policy model, a reference model, and a source of preferences, but optimization is framed as supervised learning on preference pairs.

Finally, Grouped Relative Preference Optimization (GRPO) (Shao et al., 2024) operates on groups of candidates per input instead of individual samples. For each prompt X , the policy produces a small set of responses $\{Y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | X)$, which are scored by a reward model or verifier to obtain scalar rewards $r_i = r(X, Y_i)$. These scores are then normalized within the group so that the update depends on how good each response is relative to the others for the same input, for example via

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r}, \quad \bar{r} = \frac{1}{G} \sum_{j=1}^G r_j, \quad \sigma_r = \sqrt{\frac{1}{G} \sum_{j=1}^G (r_j - \bar{r})^2}.$$

The normalized quantity $\hat{A}_i$ plays the role of an advantage for candidate Y_i : it is positive if Y_i scores above the group average and negative if it scores below. The term $\bar{r}$ is the mean reward across the G candidates for the same prompt, and acts as a baseline so that updates depend on relative quality rather than absolute reward scale. The term σ_r is the standard deviation of rewards within the group.

The policy is updated to increase the probability of above-average responses and decrease the probability of below-average ones, using a PPO-style clipped objective over tokens and a regularization term that keeps it close to a reference model:

$$J_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|Y_i|} \sum_{t=1}^{|Y_i|} \min \left(\rho_{i,t}(\theta) \hat{A}_i, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta D_{\text{KL}}(\pi_{\theta}(\cdot | X) \| \pi_{\text{ref}}(\cdot | X)) \right],$$

where $\rho_{i,t}(\theta) = \frac{\pi_{\theta}(Y_{i,t} | X, Y_{i,<t})}{\pi_{\theta_{\text{old}}}(Y_{i,t} | X, Y_{i,<t})}$ is the PPO likelihood ratio. Liu et al. (2025) show that naive choices in GRPO, in particular some normalization and weighting steps, can introduce biases. Therefore, they propose Dr. GRPO, a corrected variant that modifies the normalization and weighting so that the update better matches the intended policy-gradient objective while retaining the practical benefits of group-wise comparisons.

3 Related Work

This section reviews prior work on handling Portuguese language varieties and situates our approach in the broader literature. Section 3.1 surveys studies on identification of European and Brazilian Portuguese, from early n -gram-based classifiers on silver labels to recent cross-domain transformer models. Section 3.2 focuses on translation to EP and BP, tracing the evolution from Statistical Machine Translation (SMT) to Neural Machine Translation (NMT) and LLM-based systems tailored to European Portuguese. Section 3.3 broadens the scope to related work on language-variety and dialect modeling beyond Portuguese. Finally, Section 3.4 reviews RL and preference-feedback approaches for language models.

3.1 Portuguese Variety Classification

Numerous works have addressed EP and BP identification. Early studies relied on silver-labeled data and n -gram features, reporting high effectiveness on heuristic labels (Da Silva and Lopes, 2006; Zampieri and Gebre, 2012a; Castro et al., 2016). However, when evaluated on the later gold-labeled DSL-TL corpus (Zampieri et al., 2024), results dropped notably, underscoring the challenge of true variety discrimination and the importance of human annotation. Zampieri and Gebre (2012b) propose one of the first automatic systems for Portuguese variety identification, using character and word n -grams to discriminate between European and Brazilian Portuguese. The models achieve high accuracy in this setting, showing that simple n -gram models already capture many orthographic and lexical cues that separate the two varieties.

More recently, Sousa et al. (2025b) propose a cross-domain Portuguese language-variety identifier. The authors compile PtBrVarId across six domains (journalistic, web, social media, literature, legal, politics) and study Transformer versus n -gram baselines. Their best setting fine-tunes BERTimbau (Souza et al., 2020) with a two-step cross-domain protocol and applies delexicalization (masking named entities and POS-selected tokens) during training. Evaluated on the DSL-TL and FRMT datasets, partial delexicalization improves generalization and the BERT-based classifier outperforms the n -gram and Naive Bayes baselines, indicating that cross-domain robustness benefits from transformer representations and from reducing reliance on domain-specific lexical cues via delexicalization.

3.2 Portuguese Variety Translation

Research on EP-BP transformation has progressed from statistical methods to neural approaches and, more recently, to LLM-based systems. Early work relied on SMT: Marujo et al. (2011) adapt BP texts to EP using TED-style phrase tables in a phrase-based SMT system, showing that even closely related varieties benefit from dedicated adaptation.

Subsequent work marked a shift toward neural methods. Costa-jussà et al. (2018) present an NMT system specifically targeting national varieties of Portuguese. Their recurrent encoder-decoder model

with attention is trained in both directions and yields BLEU improvements over phrase-based SMT, alongside a preference for neural outputs in human evaluation. The authors also situate their contribution within broader efforts on language-variety translation and multilingual NMT, highlighting that multilingual encoders can capture cross-variety regularities.

More recent studies explore modern encoder-decoder and decoder-only architectures. [Sanches et al. \(2024\)](#) curate the 500-sentence Gold Collection dataset and fine-tune mBART-50 and M2M100 on subtitles and TED transcripts. They report both automatic and human evaluations, and compare against ChatGPT-3.5. Although their best mBART variant is competitive, it generally does not surpass GPT-3.5, underscoring the strength of large instruction-following models even without task-specific fine-tuning. The same study also reviews a broader spectrum of EP-BP adaptation methods, including rule-based systems augmented with masked language models, fine-tuned multilingual NMT systems, and GPT-4-based approaches, and reports mixed evidence on whether generic LLMs or targeted fine-tuned models are preferable for very closely related varieties.

To the best of our knowledge, the current state of the art is represented by [Sousa et al. \(2025a\)](#), who introduce PTradutor, the first open-source translation model explicitly tailored to European Portuguese. Their system reframes EP-targeted MT as instruction-conditioned language modeling on a decoder-only architecture. Evaluated on FRMT and NTREX, PTradutor surpasses existing open baselines and approaches strong closed-source systems such as GPT-4, demonstrating that decoder-only LLMs can serve as competitive NMT systems for language varieties when properly instruction-tuned.

3.3 Classification and Translation Outside Portuguese

Beyond Portuguese, transfer learning from pre-trained NMT models is a standard approach for closely related or low-resource language pairs. [Adebara and Abdul-Mageed \(2021\)](#) fine-tune pre-trained Transformer MT models for pairs such as Catalan-Spanish, Portuguese-Spanish, and French-Bambara, showing that transfer learning substantially outperforms training from scratch and that tokenization choices further affect performance.

Related findings also appear on translation between national varieties. Using pre-trained multilingual NMT backbones, [Lakew et al. \(2018\)](#) study translation into closely related varieties such as Canadian-European French and Indonesian-Malay within a multilingual NMT framework. Their results indicate that cross-variety distinctions can be learned effectively even with relatively small amounts of supervised data, highlighting the value of transfer learning for fine-grained linguistic variation.

Large instruction-following models have likewise become competitive baselines in machine translation. A preliminary evaluation by [Jiao et al. \(2023\)](#) reports strong ChatGPT performance for well-resourced European languages, but substantially weaker results under domain noise and in low-resource settings.

In parallel to translation-focused work, several efforts emphasize evaluation resources and benchmarks that frame dialect and language-variety processing primarily as classification. [Faisal et al. \(2024\)](#) introduce DIALECTBENCH, a benchmark organized around structured language clusters, defined as groups of related language varieties that share a common linguistic root (e.g., Arabic dialects or Swiss German dialects). DIALECTBENCH aggregates resources for these clusters into a unified evaluation framework. Building on this benchmark, [Faisal and Anastasopoulos \(2025\)](#) define seven tasks, including POS tagging, dialect identification, sentiment analysis, topic classification, natural language inference, machine reading comprehension, and extractive QA, spanning dozens of dialects and language varieties. Comparing supervised fine-tuning of encoder models (mBERT, XLM-R) to k -shot in-context learning with LLMs, they find that LLMs perform strongly on comprehension-style tasks but lag behind encoder-based models on multi-class dialect identification.

3.4 Reinforcement Learning and Preference Feedback for Language Models

Recent work has explored RL for machine translation under bandit or preference-based feedback. [Nguyen et al. \(2017\)](#) NMT as a contextual bandit problem in which, instead of full reference translations, the system receives scalar rewards that mimic noisy human ratings. Combining an advantage actor-critic

algorithm with an encoder-decoder NMT model, they show that learning from simulated human feedback is robust to skewed, high-variance reward signals and can directly optimize corpus-level MT objectives.

RL has also been applied in dialect-sensitive translation scenarios. [Thu et al. \(2023\)](#) fine-tune Burmese dialect translation models via RL, initializing from an existing NMT system and performing policy updates guided by automatic quality metrics. Their approach improves adequacy and fluency for under-represented dialect pairs, illustrating how RL can be used not only to increase overall BLEU but also to steer translation behavior toward closely related language varieties.

More recent work integrates reward models and preference models explicitly into MT pipelines. [Ramos et al. \(2024\)](#) investigate quality-estimation metrics trained on human annotations as learned reward models for multiple stages, including data filtering, RL-based training, and inference-time reranking. They find that combining RL training with reranking guided by these learned quality models yields substantial gains over strong supervised baselines. [Xu et al. \(2024\)](#) study translation preference modeling with RL from human feedback (RLHF) using a cost-effective two-step strategy. First, the authors train a reward model as a binary classifier that distinguishes human expert translations from outputs of a supervised machine translation system using existing parallel data. Then, the authors use this reward model to provide scalar feedback in a PPO-style policy optimization step. In this way, the reward model captures systematic deficiencies of the system relative to human translations without requiring expensive pairwise preference collection for each input, and its signal can guide improvements even for directions lacking explicit human-comparison data.

Beyond machine translation, there is growing interest in AI-generated feedback as an alternative to direct human supervision for preference-based RL. [Tu et al. \(2024\)](#) introduce RL-SaLLM-F, an online method that replaces hand-crafted reward functions or scripted teachers with feedback from a large language model. In standard RL terminology, an environment trajectory is the sequence of states, actions, and (true) rewards obtained by an agent during interaction. RL-SaLLM-F abstracts each trajectory into a natural-language description and uses an LLM in two roles: it produces preference labels over pairs of trajectory descriptions and generates self-augmented imagined trajectories that improve on the observed ones. A double-check mechanism is employed to reduce randomness in these LLM judgments. This LLM-based feedback eliminates the need for privileged environment rewards or real-time human oversight while still enabling effective preference-based policy optimization.

Similarly, [Lee et al. \(2024\)](#) RLHF with reinforcement learning from AI feedback (RLAIF) across summarization and dialogue benchmarks. The authors show that reward models trained on preferences generated by a strong LLM can match or slightly surpass the downstream performance of reward models trained on human comparisons, and they further propose a direct-RLAIF variant that queries an off-the-shelf LLM for rewards without a separate reward-model training phase.

More recently, [Zhang et al. \(2025\)](#) survey RL methods for Large Reasoning Models (LRMs) and provide a useful taxonomy that also applies to preference-driven machine translation. The authors distinguish verifiable from learned (judge-based) rewards, and review policy-optimization objectives ranging from PPO to critic-free preference losses such as DPO and group-based variants such as GRPO. The authors highlight open questions, including whether RL enhances existing models or enables qualitative discovery beyond SFT, and how different reward types affect this trade-off.

4 Corpus Construction

Compiling corpora suitable for language-variety NLP tasks is non-trivial. To support the objectives of this work, we construct a unified corpus from four datasets, selected to serve complementary roles in variant classification and translation. Some resources are used in both tasks, while others are used exclusively for classification. We organize the corpus into two instances: a train instance and a test instance. The training instance aggregates OPUS OpenSubtitles, PtBrVarId (train and validation splits), and the FRMT dev split. The testing instance aggregates the PtBrVarId test split, the FRMT test split, and the Gold Collection dataset. When available, we preserve dataset-specific metadata fields (e.g., *bucket* in the FRMT dataset) to enable stratified analyses. Table 1 summarizes the resulting structure of the train and test instances.

OPUS OpenSubtitles is an open collection of sentence-aligned movie and TV series subtitles, developed to support machine translation and cross-linguistic studies. FRMT is a benchmark for few-shot machine translation that includes produced translations from English into EP and BP. PtBrVarId is a cross-domain language-variety identification resource designed to discriminate between EP and BP, spanning six textual domains: Journalistic, Legal, Politics, Web, Social Media, and Literature. Finally, the Gold Collection is a manually curated evaluation set for Brazilian-to-European Portuguese adaptation comprising 500 sentences across five topics, each paired with two distinct reference rewrites. Specifically, the dataset covers Science, Legislative, Literature, Social Media, and Wiki domains.

The remainder of this section is organized as follows. Section 4.1 presents the OPUS OpenSubtitles preprocessing pipeline, Section 4.2 describes the PtBrVarId preprocessing pipeline, and Section 4.3 characterizes the resulting corpus.

Table 1: Database structure and column description for train and test instances.

Column	Description
dataset	Source tag: OpenSubs PtBrVId FRMT Gold
bucket	Domain/bucket (when available)
theme	Optional theme/topic (when available)
text_pt_br	Brazilian Portuguese text (or NULL if not applicable)
text_pt_pt	European Portuguese text (or NULL if not applicable)
ref_pt_pt_manual	Manual EP reference (for Gold data)
ref_pt_pt_deepl	DeepL EP reference (for Gold data)

4.1 OpenSubtitles Preprocessing Pipeline

Since OPUS OpenSubtitles provides the majority of the data used in this work, it is essential to ensure that subtitle pairs are well aligned and that both sides convey the same information. To this end, we apply a normalization pipeline that restores sentence boundaries and reduces formatting artifacts that interfere with alignment. Specifically, we normalize subtitle lines, drop speaker labels, fix spacing around punctuation, remove dialogue dashes at line starts and after punctuation, convert all-caps lines to sentence case, and trim wrapping quotation marks.

The subtitles in the OpenSubtitles dataset were aligned automatically based on temporal overlap between subtitle frames (Tiedemann, 2009). However, we found many instances where the aligned segments did not correspond cleanly across both EP and BP variants. A common issue is that a sentence continuation appears in the following row on one side (e.g., EP) but remains in the current row on the other (e.g., BP). To mitigate this, we implement a sentence merging procedure that detects likely continuation boundaries and merges adjacent lines when appropriate. Concretely, we merge when the previous line ends with a comma, semicolon, colon, dash, and the following line begins with a lowercase letter. Conversely, we avoid merging when the previous line ends with strong punctuation (., ?, !, . . .) and the next line begins with an uppercase letter. After merging, we capitalize the first alphabetical character of any sentence that follows a strong stop.

Following normalization and merging, we apply a coarse outlier detection procedure based on sentence-length ratios. For each EP-BP pair, we compute the character length on each side and use the ratio

$$\text{ratio} = \frac{\text{len}_{PT}}{\text{len}_{BR}},$$

to filter outliers. The BR and PT sentence lengths are strongly correlated, with a Pearson coefficient of $r = 0.963$. Length ratios have a median of 1.00 and a mean of 1.03. In the OPUS corpus, approximately 80.40% of pairs fall in the stricter [0.8, 1.25] interval and 98.46% fall in the looser [0.5, 2.0] interval. Figure 1 illustrates the sentence-length scatter (for a sample of pairs) and the conservative outlier bounds.

We treat pairs that violate the conservative band $0.5 \leq \text{ratio} \leq 2.0$ as hard outliers and remove them from the corpus. This filter removes 161,854 subtitle pairs, corresponding to 1.54% of the OPUS data.

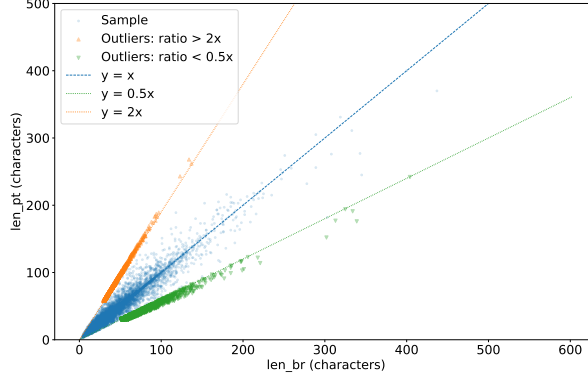


Figure 1: Sentence-length correlation in the OpenSubtitles dataset. The dashed line is $y = x$ and dotted lines mark $y = 0.5x$ and $y = 2x$. A small sample of flagged outliers is overlaid for legibility.

In addition to global length mismatches, we observed cases where both sides share well-aligned material at the beginning and end, but one side contains an unmatched block in the middle. To target these misalignments, we introduce a second filter based on token-level links produced by SimAlign (Sabet et al., 2021). SimAlign constructs a token-to-token similarity matrix using multilingual encoders such as mBERT (Pires et al., 2019) or XLM-R (Conneau et al., 2020), and then extracts hard alignment links using one of three strategies. Argmax links two tokens only when they are mutual nearest neighbors in the similarity matrix. IterMax applies Argmax iteratively: after each pass, newly created links are locked by removing their row/column from consideration, and the remaining scores of tokens aligned on only one side are slightly reduced. This iterative cleanup can reveal additional links that were previously overshadowed. As a result, IterMax is the only strategy that naturally permits one-to-many relations when sufficient evidence remains. Finally, the Match method enforces a one-to-one alignment by solving a maximum-weight bipartite matching problem. Below we illustrate how these strategies link tokens between example EP and BP sentences.

BR : Ele está correndo

PT : Ele está a correr

Tokens: [Ele][está][correndo] [Ele][está][a][correr]

Argmax: Ele $\leftrightarrow$ Ele, está $\leftrightarrow$ está, correndo $\leftrightarrow$ {a or correr}

IterMax: Ele $\leftrightarrow$ Ele, está $\leftrightarrow$ está, correndo $\leftrightarrow$ a, correndo $\leftrightarrow$ correr

Match: Ele $\leftrightarrow$ Ele, está $\leftrightarrow$ está, correndo $\leftrightarrow$ correr

For each EP-BP pair, we run SimAlign with the IterMax strategy to obtain token-level links and then compute a base similarity score defined as the proportion of unique word types that participate in at least one link, relative to the set of unique word types across both sentences. As short sentences contain fewer informative tokens, we apply a length-dependent decision threshold. Concretely, rows with eight tokens or fewer are evaluated against a minimum similarity of 0.30, rows with twenty-eight tokens or more are evaluated against a 0.70 threshold, and the threshold increases linearly between these endpoints. Pairs whose similarity falls below the length-appropriate threshold are flagged and removed from the working table of subtitle pairs.

Together, normalization, sentence merging, and the two-stage filtering procedure substantially reduce subtitle-specific noise and misalignment. We next describe the preprocessing applied to PtBrVarId, which poses different challenges due to its heterogeneous domains and web-derived content.

Dataset	Train	Valid	Test	Total
OpenSubs	10,347,883	0	0	10,347,883
PtBrVarId	2,959,559	29,680	1,450	2,990,689
FRMT	3,915	20	3,987	7,922
Gold	0	0	500	500
Total	13,311,357	29,700	5,937	13,346,994

Table 2: Number of rows per dataset and split in the final corpus.

4.2 PtBrVarId Preprocessing Pipeline

PtBrVarId contributes a substantial portion of the sentences used in this work. Therefore, we apply an extensive preprocessing pipeline to maximize data quality. The version released by [Sousa et al. \(2025b\)](#) does not preserve accents, so we instead rely on the raw version of the dataset. Most of the preprocessing steps below follow the procedure described by [Sousa et al. \(2025b\)](#).

We begin by dropping empty rows and applying `jusText` ([Belica, 2025](#)), a boilerplate-removal library that classifies textual blocks using features such as length and stop-word density. As noted by [Sousa et al. \(2025a\)](#), this step is intended to filter low-quality content and has the largest impact on the Social Media domain, removing approximately 2.4 million documents. Figure 2 (right) reports the number of documents removed by `jusText` per domain.

Since we work with the original PtBrVarId data, we additionally remove hashtags and URLs, normalize quotation marks, drop speaker labels, remove duplicates, and capitalize sentence-initial tokens. The pipeline concludes with an interquartile-range (IQR) filter to retain texts whose lengths fall within

$$[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}].$$

Overall, the full pipeline removes approximately 6 million documents from the 9 million available in the raw PtBrVarId dataset. Figure 2 (left) summarizes the cumulative removals across the successive stages of our pipeline.

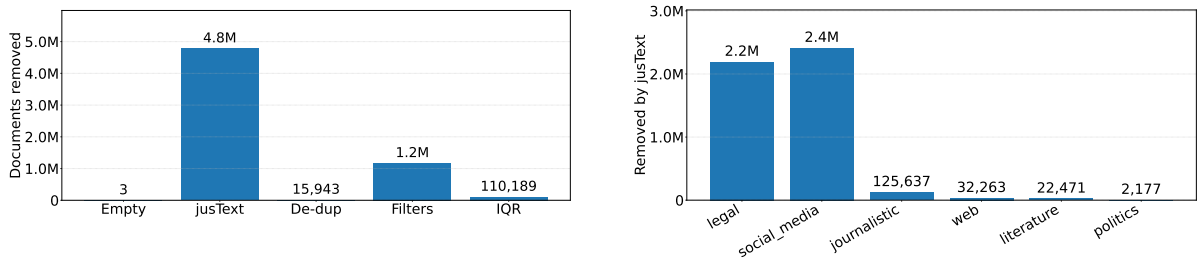


Figure 2: Left: Removals in PtBrVarId in each step of the preprocessing pipeline. Right: JusText removals by domain in PtBrVarId.

4.3 Characterization of the Resulting Corpus

After preprocessing the data, we report corpus statistics over the train, validation, and test splits. In total, the combined corpus contains 13,346,994 rows: 13,311,357 in the training split, 29,700 in validation, and 5,937 in test. Table 2 summarizes the distribution of rows across the four resources. Most training data originates from OpenSubtitles, while PtBrVarId contributes approximately 3 million classification documents across the three splits. FRMT is also present in all splits but remains comparatively small (approximately 7,922 rows in total). The Gold Collection is restricted to the test split and contributes 500 rows.

Split	Variety	n	Mean	Median	P90	P99	Max
Train	BR	10,684,022	53	32	86	482	4,087
Train	PT	12,976,311	136	39	413	1,158	4,523
Valid	BR	3,330	452	389	725	1,398	3,805
Valid	PT	26,390	516	409	974	1,456	4,045
Test	BR	3,288	162	138	276	603	3,109
Test	PT	3,888	279	189	622	1,293	3,947

Table 3: Character-length statistics per split and Portuguese variant.

Split	Total tokens (BR)	Mean tokens (BR)	Total tokens (PT)	Mean tokens (PT)
Train	174,355,295	13.10	516,309,250	38.79
Valid	441,586	14.87	3,853,849	129.76
Test	155,011	33.96	308,960	67.68

Table 4: Approximate token counts per split and variety, using the Qwen-3 0.6B tokenizer.

To assess whether segment-length distributions are representative, we compute character-based statistics over all non-empty segments for each Portuguese variety (see Table 3). In the training split, segments are relatively short and less variable, reflecting the dominance of OpenSubtitles. In this split, EP segments are, on average, roughly two to three times longer than BP segments. European Portuguese also exhibits a heavier tail: high-percentile values extend well beyond those of BP, consistent with the presence of more EP-only material from PtBrVarId in the training data.

In contrast, validation and test segments are substantially longer on average, since these splits are dominated by FRMT and PtBrVarId rather than subtitles. The medians of characters are significantly larger and the upper quantiles approach or exceed one thousand characters on both sides, indicating a pronounced long tail in the evaluation data.

Finally, using the Qwen-3 0.6B tokenizer, we estimate that the training split contains approximately 174M BR tokens and 516M PT tokens, with mean segment lengths of 13 and 39 tokens, respectively (see Table 4). Although the validation split is much smaller in absolute size, PT segments are longer on average (around 130 tokens). The test split lies between the two, with mean lengths of roughly 34 tokens (BP) and 68 tokens (EP).

5 Thesis Proposal

This section states the central hypothesis of this work and outlines the steps required to validate it. Section 5.1 presents the general proposal that this study aims to investigate. The subsequent subsections elaborate the topic in more detail: Section 5.2 describes planned refinements to corpus construction and filtering, Section 5.3 outlines the model-training and adaptation plan, and Section 5.4 discusses preference scoring and reinforcement-learning-based extensions. Finally, Section 5.5 reports preliminary experiments and documents the training setup (Section 5.5.1) and evaluation protocol (Section 5.5.2) used so far.

5.1 General Proposal

This work plans to investigate whether it is possible to develop a variety-aware model component that can be reused to help larger language models reliably distinguish European Portuguese from Brazilian Portuguese, and to follow explicit EP/BP preferences in both classification and rewriting settings, either through further fine-tuning or by using the component as a test-time scorer for selection and re-ranking.

Validating this hypothesis requires a consolidated and well-controlled corpus that exposes systematic EP-BP contrasts and supports both labeled identification and bidirectional rewriting. Moreover, it requires a robust preprocessing and filtering pipeline that can be iteratively refined to reduce noise and misalignment while preserving useful variety-specific signal. On the modeling side, validation requires systematic experimentation with multiple decoder-only models and adaptation strategies, in order to identify configurations that best generalize across domains under the available compute constraints. Finally, validation requires an evaluation protocol that reports quantitative performance for each variety and enables qualitative error analysis to diagnose failures.

5.2 Planned Refinements to Corpus and Filters

Given the importance of data quality for the next stages of model development, we plan to refine the corpus and its filtering procedures. For OPUS OpenSubtitles, additional subtitle material will be incorporated, and the preprocessing pipeline will be extended to cover a broader set of language pairs. The sentence-length ratio filter will be reassessed by exploring alternative threshold settings, with the objective of removing clearly mismatched pairs while avoiding the exclusion of well aligned subtitles.

The SimAlign-based filter will likewise be revised. Experiments will be conducted with alternative multilingual encoders and alignment extraction strategies, and the similarity thresholds governing pair retention will be tuned accordingly. Once the revised filters have been finalized, the descriptive statistics will be updated to reflect the refined corpus.

5.3 Model Training and Adaptation Plan

In parallel with the ongoing data refinement, we aim to expand the range of models and training strategies under consideration. The initial Qwen-based system can be maintained as a baseline, while additional decoder-only architectures from the Qwen and Gemma families will be investigated at scales that remain feasible given the available hardware. In addition, the MiniLingua model introduced in Section 2.4 may constitute a suitable alternative, particularly given its potential relevance to the multilingual setting addressed in this work.

For each candidate model, the focus will remain on the two principal tasks of this work. The present Qwen-based system is trained in a multitask setting that jointly optimizes both tasks. One may consider to include a systematic comparison with single-task training. This comparison is intended to determine whether task separation yields better results.

A further central consideration concerns the choice between PEFT and full fine-tuning. LoRA and related parameter-efficient methods will be evaluated against full weight updates for a selected subset of model sizes. The evaluation will account for performance on both tasks as well as computational requirements, including memory consumption.

In the specific case of PEFT techniques, the LoRA rank, scaling factor, and target modules will be varied to assess their effect on model performance. For full fine-tuning, training will begin with conservative learning rates and regularization, and overfitting will be closely monitored. The results of these experiments will guide the selection of the final models and identify configurations that provide an appropriate trade-off between output quality and computational cost.

5.4 Variety Preference Scoring and RL-Based Adaptation

In addition to supervised fine-tuning, the best-performing model will be repurposed as a variety preference scorer. Given an instruction specifying the Portuguese variant, a source sentence, and a candidate output, the model assigns the candidate a scalar score (its conditional log-likelihood under the instruction and source).

At inference time, we use this score as a proxy for variety consistency: higher-scoring outputs are treated as more compatible with the requested variety, while lower-scoring outputs are treated as less aligned.

We plan to exploit these scores in two complementary ways. First, they support selection and re-ranking: given multiple candidates (e.g., sampled from an external model), the scorer selects the output that best matches the target variety. Second, they can serve as a fixed reward signal for RL-based adaptation, by using the scorer as a generative reward model within a procedure such as Dr. GRPO (Liu et al., 2025). In this setting, groups of candidate responses are scored, probability-based rewards are derived to reflect the degree to which each output satisfies the requested variety, and a larger decoder-only policy model is updated to increase the expected reward.

The overarching objective is to obtain systems that reliably prioritize European Portuguese when requested, without degrading performance on Brazilian Portuguese. The final work will therefore compare models trained with and without this preference signal.

5.5 Preliminary Results

This subsection summarizes the work completed to date and reports initial results. So far, the effort has focused on establishing a reliable dataset and training an initial model that jointly targets the central aims of this study.

As described in Section 4, we built an initial corpus by aggregating OPUS OpenSubtitles, FRMT, PtBrVarId, and the Gold Collection. This effort began with the development of two dataset-specific preprocessing pipelines, one for OPUS OpenSubtitles and one for PtBrVarId, which produced the first cleaned and consolidated version of the corpus.

Based on this cleaned dataset, an initial decoder-only model from the Qwen3 family was fine-tuned using LoRA adapters in a multitask setting that combines variety classification with bidirectional rewriting between EP and BP. The remainder of this subsection describes the training formulation and optimization setup, and reports quantitative and qualitative evaluation on the test split.

5.5.1 Training

We use the Qwen3-0.6B model, which comprises 28 Transformer blocks, with 16 query heads and 8 key-value heads per layer (Yang et al., 2025). Each block follows the standard decoder-only architecture, consisting of masked self-attention followed by a feed-forward network, with residual connections and RMSNorm applied throughout. Rotary positional embeddings and grouped-query attention are employed as described in Section 2.

Relative to earlier Qwen generations, Qwen3 is trained on a substantially larger and more diverse dataset, amounting to 36 trillion tokens spanning 119 languages and dialects, and incorporating high-quality material from coding, STEM (Science, Technology, Engineering, and Mathematics), reasoning-oriented data, books, multilingual web sources, and synthetic corpora. Tokenization relies on the original Qwen tokenizer (Bai et al., 2023), thereby preserving compatibility with the pre-trained vocabulary and the associated chat template.

Translation (pt-BR $\rightarrow$ pt-PT)	Variant classification
<code>< im_start >system</code> SYSTEM_TRANSLATION <code>< im_end ></code> <code>< im_start >user</code> “Converte o seguinte texto de português do Brasil para português europeu” “Texto: {source}” <code>< im_end ></code> <code>< im_start >assistant</code> {target_ptpt} <code>< im_end ></code>	<code>< im_start >system</code> SYSTEM_CLASSIFICATION <code>< im_end ></code> <code>< im_start >user</code> “Qual é a variante de português desta frase? Responde apenas com ‘Português do Brasil’ ou ‘Português Europeu’.” “Frase: {source}” <code>< im_end ></code> <code>< im_start >assistant</code> {“pt-BR” “pt-PT”} <code>< im_end ></code>

Table 5: Chat-template structure of our training examples for BP to EP translation and variant-classification tasks. The placeholders {source} and {target} are filled from the corpora described in Section 4.

We then formulate EP-BP both translation and variety identification tasks as a unified supervised instruction-following problem:

- **Translation (BP $\rightarrow$ EP):** For parallel sentence pairs with both BP and EP sides, we construct instances of the form “*Converte o seguinte texto de português do Brasil para português europeu*”, using the BP sentence as input and the EP sentence as the target output.
- **Translation (EP $\rightarrow$ BP):** For parallel sentence pairs with both BP and EP sides, we construct instances of the form “*Converte o seguinte texto de português europeu para português do Brasil*”, using the EP sentence as input and the BP sentence as the target output.
- **Classification (EP vs. BP):** For sentences with a single variety label, we prompt the model to output only the variety, using the instruction “*Qual é a variante de português desta frase? Responde apenas com ‘Português do Brasil’ ou ‘Português Europeu’.*”, and supervise the model on the corresponding label (pt-BR or pt-PT).

The chat templates used for the translation and classification tasks are presented in Table 5. Each template follows a common structure comprising a system message that specifies the model’s role and the target variety, a user message that provides the source sentence together with the instruction, and an assistant message that contains the desired output. For translation, two templates are used, depending on the direction of translation. For classification, the template elicits a single variety label. Each system-user-assistant triplet is converted into the Qwen chat format and used to train a causal language model on the resulting text. The training objective is next-token prediction over the full conversational sequence, such that the model learns to generate all tokens in the triplets conditioned on the preceding tokens. Padding tokens are excluded from the cross-entropy loss.

In the experiments, we use the Qwen3-0.6B chat tokenizer with left padding and a maximum context length of 2048 tokens. Model adaptation is carried out using LoRA. Specifically, LoRA modules are inserted into the attention and MLP projection layers, with rank $r = 32$, scaling factor $\alpha = 32$, and dropout $p = 0.05$. The underlying Qwen parameters are kept frozen, and only the LoRA adapter parameters are updated during training.

Optimization is conducted with the HuggingFace Trainer, which employs AdamW with a linear warmup and linear decay schedule. The learning rate is set to 1×10^{-4} with weight decay 0.01. Training uses a per-device batch size of 2 with gradient accumulation over 4 steps, at a context length of 2048 tokens. A maximum of 1,300,000 training steps is specified, although training may terminate earlier via early stopping. Validation is performed every 10,000 steps, and early stopping with a patience of 3 is applied to halt training when the validation loss ceases to improve.

We next outline the evaluation setup and metrics used to assess both rewriting quality and classification performance on the test split.

5.5.2 Evaluation

Evaluation is conducted using two families of metrics. For EP-BP rewriting, we report sacreBLEU (Post, 2018). For variety classification, we report accuracy together with per-class precision and recall. In addition, macro-F1 and balanced accuracy are also included, as they are less sensitive to class priors. Balanced accuracy is computed as the mean recall over BP and EP, and macro-F1 is computed as the mean of the per-class F1 scores. Overall results for rewriting and classification are summarized in Table 6, while Table 7 reports the per-class precision, recall, and F1 scores for the classification task. Qualitative examples for classification outputs, including common error patterns, are shown in Table 8.

For the translation task, we report sacreBLEU (Post, 2018). Overall results for EP-BP rewriting are summarized in Table 6, including direction-wise scores and bucketed BLEU results (entity, lexical, random). Qualitative analysis is provided through representative examples. Table 9 illustrates rewritten instances in which the model frequently copies the source rather than producing variety-specific substitutions, even when the reference text contains clear variety-specific vocabulary.

Task / Split	Metric	Score
Translation (overall)	BLEU	30.14
br2pt	BLEU	30.68
pt2br	BLEU	29.60
Translation (bucket=entity)	BLEU	33.09
Translation (bucket=lexical)	BLEU	24.31
Translation (bucket=random)	BLEU	32.69
Classification (overall)	Accuracy	60.35
Classification (overall)	Balanced accuracy	56.90
Classification (overall)	Macro-F1	49.67

Table 6: Preliminary test results of balanced accuracy and macro-F1.

Class	Precision	Recall	F1	Support
pt-BR	88.28	15.57	26.47	3288
pt-PT	57.91	98.23	72.86	3888

Table 7: Classification statistics on the test split.

Input (source)	Gold	Pred.
<i>(PtBrVarId)</i> Alguns grupos já foram excluídos do PT como a “Causa Operária” e “Convergência Socialista”, de extrema esquerda. A direção do PT entendeu que eles não seguiam as orientações da legenda. Quase todas as tendências petistas foram formadas por ex-militantes de organizações clandestinas de esquerda durante o regime militar.	pt-BR	pt-PT
<i>(PtBrVarId)</i> Duarte, nos dois jogos que dirigiu a equipe no estádio Bruno José Daniel, atuou com três atacantes, num esquema um pouco mais ofensivo. Picerni prefere atuar de uma maneira mais cautelosa. Já na primeira partida que fez com a equipe conseguiu uma vitória contra a Ponte Preta, em Campinas.	pt-BR	pt-BR
<i>(PtBrVarId)</i> Porque o poeta não chama as Nymphas para que venhão aplacar os seus Prantos e clamores . . . Chama-as para que os venhão ouvir, e para que vejam a que estado o tem Reduzido o seu amor . . .	pt-PT	pt-BR

Table 8: Qualitative examples for variety classification.

Source	Reference	Output
<i>FRMT, BR→PT</i> (A família de Rousseff não tinha conhecimento sobre a participação dela em atividades clandestinas).	(A família de Rousseff não tinha conhecimento da participação dela em atividades ilícitas).	A família de Rousseff não tinha conhecimento sobre a participação dela em atividades clandestinas.
<i>FRMT, PT→BR</i> A Costa dos Murmúrios foi adaptada recentemente ao cinema por Margarida Cardoso.	A Costa dos Murmúrios foi adaptada recentemente para o cinema por Margarida Cardoso.	A Costa dos Murmúrios foi adaptada recentemente ao cinema por Margarida Cardoso.
<i>FRMT, BR→PT</i> Uma hierarquia com ônibus de seis tamanhos que se alimentam.	Uma hierarquia de autocarros de seis tamanhos controlam-se automaticamente.	Uma hierarquia com ônibus de seis tamanhos que se alimentam.

Table 9: Qualitative examples for EP-BP rewriting.

The classification statistics indicate a strong systematic bias towards the European Portuguese label. On the full test set ($n = 7171$), only 513 out of 3287 BP instances are predicted correctly, while 2774 BP instances are assigned EP. In contrast, 3816 out of 3884 EP instances are classified correctly, with only 68 misclassified as BP. As a result, EP is produced for approximately 91.9% of all inputs, yielding a BP recall of 15.6% and a EP recall of 98.2% (balanced accuracy $\approx 56.9\%$). This tendency is consistent across the bucketed subsets of the FRMT dataset.

Several factors may plausibly contribute to this behaviour on the classification task. The multi-task training mixture is not uniformly balanced across data sources and splits, and the validation distribution is skewed towards EP, which can bias both early stopping and the learned decision boundary. In addition, the classification prompt is written in a markedly European register, which may function as an implicit prior towards this variant when the model is uncertain.

Separately, there are indications that the rewriting objective may not provide sufficiently strong supervision. Many EP-BP sentence pairs in the rewriting data are identical or near-identical. Under

an autoregressive objective over chat-formatted sequences, copying the input can yield low loss, thereby reducing the incentive to learn systematic variety-specific edits. Consistent with this interpretation, qualitative inspection of the model outputs suggests frequent source copying in the rewriting task, with outputs that are identical or nearly identical to the input while diverging from the reference.

To mitigate these issues for rewriting, the next iteration might filter or down-weight translation pairs whose EP and BP content are identical or differ only trivially, so that the rewriting objective provides stronger and less ambiguous supervision. To address the classification bias, the multi-task mixture will be rebalanced and the classification loss will be reweighted to improve BP recall. The prompts will also be revised towards a more neutral register, and classification outputs will be constrained to short labels (e.g., pt-BR/pt-PT) rather than full-form responses such as “*Português do Brasil*” and “*Português Europeu*”. Finally, training will be migrated to *Axolotl* to enable a more systematic exploration of efficiency and stability choices, including packing and padding strategies, attention implementations, LoRA target modules and ranks, and effective batch sizes under the available hardware constraints.

6 Conclusions

This work aims to develop and evaluate controllable rewriting models for European and Brazilian Portuguese, with the goal of producing variety-appropriate outputs when explicitly requested. We start with the construction of a EP/BP corpus with a reproducible preprocessing pipeline, and a supervised fine-tuning experiment using a decoder-only model. Building on the strongest supervised configurations, subsequent stages will extend the empirical study by comparing models and training strategies, including full fine-tuning and parameter-efficient approaches. The work will also investigate preference-based adaptation by leveraging a variety-aware scorer for re-ranking and for optimization. Performance will be evaluated using classification accuracy alongside rewriting-quality metrics.

The work planned from mid-February to July is organized into partially overlapping phases. Given the experimental nature of the research, the timeline emphasizes objectives rather than fixed deadlines, and several activities are expected to proceed in parallel.

Phase 1: Corpus Refinement and Stabilization (Weeks 1 to 6). The first phase is dedicated to consolidating a stable corpus release. OPUS preprocessing will be extended, and filtering procedures will be revisited and validated through targeted manual inspection. The corpus will be adjusted to reduce training signals that encourage trivial copying, and the overall mixture will be rebalanced across varieties and sources so that training and validation splits provide more comparable EP/BP coverage. This phase concludes with a frozen corpus, updated descriptive statistics, and a documented preprocessing pipeline.

Phase 2: Training Pipeline and Prompt/Decoding Calibration (Weeks 4 to 10). Building on the stabilized corpus, the second phase refines both training and inference. Prompts will be revised towards a more neutral register and classification outputs will be constrained to short, unambiguous labels. Decoding and post-processing for rewriting will be calibrated to mitigate degenerate generations and to ensure that outputs better reflect the intended transformation. In parallel, the training workflow will be migrated to AXOLOTL to support systematic experimentation with efficiency and stability choices under fixed compute constraints.

Phase 3: Model and Training Strategy Comparison (Weeks 7 to 14). The third phase broadens the experimental scope to additional decoder-only backbones and compares training strategies under matched budgets. Multitask training will be evaluated against single-task alternatives in order to assess whether task coupling improves generalization. Parameter-efficient adaptation will be compared to full fine-tuning for selected configurations, with evaluation emphasizing both overall performance and balanced metrics that reflect symmetry between EP and BP.

Phase 4: Variety Preference Scoring and RL-Based Adaptation (Weeks 12 to 18). The fourth phase introduces preference-based adaptation on top of the strongest supervised models. A frozen variety-aware scorer will be validated and then used for re-ranking and for reinforcement-learning-style optimization with Dr. GRPO. The resulting models will be compared against purely supervised baselines, with particular focus on inputs that require non-trivial variety-specific rewriting.

Phase 5: Final Evaluation, Analysis, and Manuscript Writing (Weeks 15 to 24). The final phase consolidates results and completes the thesis manuscript. Final experiments will be run on held-out test sets, and performance will be summarized with standard classification and rewriting metrics, complemented by per-domain or per-bucket breakdowns and targeted qualitative analysis. The manuscript will be revised iteratively to integrate findings across all training strategies and to present the final methodological choices and conclusions.

Bibliography

- Ife Adebara and Muhammad Abdul-Mageed. Improving similar language translation with transfer learning. In Loic Barrault, Ondrej Bojar, Fethi Bougares, Rajen Chatterjee, Marta R. Costa-jussa, Christian Federmann, Mark Fishel, Alexander Fraser, Markus Freitag, Yvette Graham, Roman Grundkiewicz, Paco Guzman, Barry Haddow, Matthias Huck, Antonio Jimeno Yepes, Philipp Koehn, Tom Kocmi, Andre Martins, Makoto Morishita, and Christof Monz, editors, *Proceedings of the Sixth Conference on Machine Translation*, pages 273–278, Online, November 2021. Association for Computational Linguistics. URL <https://aclanthology.org/2021.wmt-1.27/>.
- Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. Gqa: Training generalized multi-query transformer models from multi-head checkpoints, 2023. URL <https://arxiv.org/abs/2305.13245>.
- Anna Aksenova, Boris Zverkov, Nicola Dainese, Alexander Nikitin, and Pekka Marttinen. Minilingua: A small open-source llm for european languages, 2025. URL <https://arxiv.org/abs/2512.13298>.
- Diego Alves. Information theory and linguistic variation: A study of Brazilian and European Portuguese. In Yves Scherrer, Tommi Jauhiainen, Nikola Ljubešić, Preslav Nakov, Jorg Tiedemann, and Marcos Zampieri, editors, *Proceedings of the 12th Workshop on NLP for Similar Languages, Varieties and Dialects*, pages 9–19, Abu Dhabi, UAE, January 2025. Association for Computational Linguistics. URL <https://aclanthology.org/2025.vardial-1.2/>.
- Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, Binyuan Hui, Luo Ji, Mei Li, Junyang Lin, Runji Lin, Dayiheng Liu, Gao Liu, Chengqiang Lu, Keming Lu, Jianxin Ma, Rui Men, Xingzhang Ren, Xuancheng Ren, Chuanqi Tan, Sinan Tan, Jianhong Tu, Peng Wang, Shijie Wang, Wei Wang, Shengguang Wu, Benfeng Xu, Jin Xu, An Yang, Hao Yang, Jian Yang, Shusheng Yang, Yang Yao, Bowen Yu, Hongyi Yuan, Zheng Yuan, Jianwei Zhang, Xingxuan Zhang, Yichang Zhang, Zhenru Zhang, Chang Zhou, Jingren Zhou, Xiaohuan Zhou, and Tianhang Zhu. Qwen technical report, 2023. URL <https://arxiv.org/abs/2309.16609>.
- Anabela Barreiro and Cristina Mota. Paraphrastic variance between European and Brazilian Portuguese. In Marcos Zampieri, Preslav Nakov, Nikola Ljubešić, Jörg Tiedemann, Shervin Malmasi, and Ahmed Ali, editors, *Proceedings of the Fifth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial 2018)*, pages 111–121, Santa Fe, New Mexico, USA, August 2018. Association for Computational Linguistics. URL <https://aclanthology.org/W18-3912/>.
- Michal Belica. justext: Heuristic-based boilerplate removal tool. <https://github.com/miso-belica/justext>, 2025. Version 3.0.2; accessed 2025-11-01.
- Yoshua Bengio, Réjean Ducharme, Pascal Vincent, and Christian Janvin. A neural probabilistic language model. *J. Mach. Learn. Res.*, 3(null):1137–1155, March 2003. ISSN 1532-4435.
- Dayvid Castro, Ellen Souza, and Adriano L.I. De Oliveira. Discriminating between brazilian and european portuguese national varieties on twitter texts. In *2016 5th Brazilian Conference on Intelligent Systems (BRACIS)*, pages 265–270, 2016. doi: 10.1109/BRACIS.2016.056.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale, 2020. URL <https://arxiv.org/abs/1911.02116>.
- Marta R. Costa-jussà, Marcos Zampieri, and Santanu Pal. A neural approach to language variety translation. In Marcos Zampieri, Preslav Nakov, Nikola Ljubešić, Jörg Tiedemann, Shervin Malmasi, and Ahmed Ali, editors, *Proceedings of the Fifth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial 2018)*, pages 275–282, Santa Fe, New Mexico, USA, August 2018. Association for Computational Linguistics. URL <https://aclanthology.org/W18-3931/>.

- Sonia Maria Lazzarini Cyrino. Algumas diferenças entre o português brasileiro e o português europeu e sua relação com a mudança sintática no português brasileiro. *Signum: Estudos da Linguagem*, 4(1): 95–112, jul. 2001. doi: 10.5433/2237-4876.2001v4n1p95. URL <https://ojs.uel.br/revistas/uel/index.php/signum/article/view/4366>.
- Joaquim Ferreira Da Silva and Gabriel Pereira Lopes. Identification of document language is not yet a completely solved problem. In *2006 International Conference on Computational Intelligence for Modelling Control and Automation and International Conference on Intelligent Agents Web Technologies and International Commerce (CIMCA'06)*, pages 212–212, 2006. doi: 10.1109/CIMCA.2006.117.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding, 2019. URL <https://arxiv.org/abs/1810.04805>.
- Fahim Faisal and Antonios Anastasopoulos. Testing the boundaries of llms: Dialectal and language-variety tasks. 2025. URL <https://par.nsf.gov/biblio/10609330>.
- Fahim Faisal, Orevaoghene Ahia, Aarohi Srivastava, Kabir Ahuja, David Chiang, Yulia Tsvetkov, and Antonios Anastasopoulos. DIALECTBENCH: An NLP benchmark for dialects, varieties, and closely-related languages. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14412–14454, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.777. URL <https://aclanthology.org/2024.acl-long.777/>.
- Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for nlp, 2019. URL <https://arxiv.org/abs/1902.00751>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.
- Frederick Jelinek. Statistical methods for speech recognition. 1997. URL <https://api.semanticscholar.org/CorpusID:12495425>.
- Zixuan Jiang, Jiaqi Gu, Hanqing Zhu, and David Z. Pan. Pre-rmsnorm and pre-crmsnorm transformers: Equivalent and efficient pre-ln transformers, 2023. URL <https://arxiv.org/abs/2305.14858>.
- Wenxiang Jiao, Wenxuan Wang, Jen-Tse Huang, Xing Wang, and Zhaopeng Tu. Is chatgpt a good translator? yes with gpt-4 as the engine. 2023. URL <https://api.semanticscholar.org/CorpusID:257631519>.
- Surafel Melaku Lakew, Aliia Erofeeva, and Marcello Federico. Neural machine translation into language varieties. In Ondřej Bojar, Rajen Chatterjee, Christian Federmann, Mark Fishel, Yvette Graham, Barry Haddow, Matthias Huck, Antonio Jimeno Yepes, Philipp Koehn, Christof Monz, Matteo Negri, Aurélie Névél, Mariana Neves, Matt Post, Lucia Specia, Marco Turchi, and Karin Verspoor, editors, *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 156–164, Brussels, Belgium, October 2018. Association for Computational Linguistics. doi: 10.18653/v1/W18-6316. URL <https://aclanthology.org/W18-6316/>.
- Tor Lattimore and Csaba Szepesvári. *Bandit Algorithms*. Cambridge University Press, 2020.
- Harrison Lee, Samrat Phatale, Hassan Mansoor, Thomas Mesnard, Johan Ferret, Kellie Lu, Colton Bishop, Ethan Hall, Victor Carbune, Abhinav Rastogi, and Sushant Prakash. Rlaif vs. rlhf: Scaling reinforcement learning from human feedback with ai feedback, 2024. URL <https://arxiv.org/abs/2309.00267>.

- Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding r1-zero-like training: A critical perspective, 2025. URL <https://arxiv.org/abs/2503.20783>.
- Luis Marujo, Nuno Grazina, Tiago Luis, Wang Ling, Luisa Coheur, and Isabel Trancoso. BP2EP - adaptation of Brazilian Portuguese texts to European Portuguese. In Mikel L. Forcada, Heidi Depraetere, and Vincent Vandeghinste, editors, *Proceedings of the 15th Annual Conference of the European Association for Machine Translation*, Leuven, Belgium, May 30–31 2011. European Association for Machine Translation. URL <https://aclanthology.org/2011.eamt-1.19/>.
- Tomas Mikolov, Martin Karafiát, Luká Burget, Jan Honza ernocký, and Sanjeev Khudanpur. Recurrent neural network based language model. In *Interspeech*, 2010. URL <https://api.semanticscholar.org/CorpusID:17048224>.
- Khanh Nguyen, Hal Daumé III, and Jordan Boyd-Graber. Reinforcement learning for bandit neural machine translation with simulated human feedback, 2017. URL <https://arxiv.org/abs/1707.07402>.
- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback, 2022. URL <https://arxiv.org/abs/2203.02155>.
- Jonas Pfeiffer, Aishwarya Kamath, Andreas Rücklé, Kyunghyun Cho, and Iryna Gurevych. Adapter-fusion: Non-destructive task composition for transfer learning, 2021. URL <https://arxiv.org/abs/2005.00247>.
- Telmo Pires, Eva Schlinger, and Dan Garrette. How multilingual is multilingual bert?, 2019. URL <https://arxiv.org/abs/1906.01502>.
- Matt Post. A call for clarity in reporting BLEU scores. In *Proceedings of the Third Conference on Machine Translation: Research Papers*, pages 186–191, Belgium, Brussels, October 2018. Association for Computational Linguistics. URL <https://www.aclweb.org/anthology/W18-6319>.
- Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2025. URL <https://arxiv.org/abs/2412.15115>.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model, 2024. URL <https://arxiv.org/abs/2305.18290>.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer, 2023. URL <https://arxiv.org/abs/1910.10683>.
- Miguel Ramos, Patrick Fernandes, António Farinhas, and Andre Martins. Aligning neural machine translation models: Human feedback in training and inference. In Carolina Scarton, Charlotte Prescott, Chris Bayliss, Chris Oakley, Joanna Wright, Stuart Wrigley, Xingyi Song, Edward Gow-Smith, Rachel Bawden, Víctor M Sánchez-Cartagena, Patrick Cadwell, Ekaterina Lapshinova-Koltunski, Vera Cabar-rão, Konstantinos Chatzitheodorou, Mary Nurminen, Diptesh Kanojia, and Helena Moniz, editors, *Proceedings of the 25th Annual Conference of the European Association for Machine Translation (Volume 1)*, pages 258–274, Sheffield, UK, June 2024. European Association for Machine Translation (EAMT). URL <https://aclanthology.org/2024.eamt-1.22/>.

- Parker Riley, Timothy Dozat, Jan A. Botha, Xavier Garcia, Dan Garrette, Jason Riesa, Orhan Firat, and Noah Constant. Frmt: A benchmark for few-shot region-aware machine translation, 2023. URL <https://arxiv.org/abs/2210.00193>.
- R. Rosenfeld. Two decades of statistical language modeling: where do we go from here? *Proceedings of the IEEE*, 88(8):1270–1278, 2000. doi: 10.1109/5.880083.
- Masoud Jalili Sabet, Philipp Dufter, François Yvon, and Hinrich Schütze. Simalign: High quality word alignments without parallel training data using static and contextualized embeddings, 2021. URL <https://arxiv.org/abs/2004.08728>.
- João Sanches, Rui Ribeiro, and Luísa Coheur. From brazilian portuguese to european portuguese, 2024. URL <https://arxiv.org/abs/2408.07457>.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL <https://arxiv.org/abs/1707.06347>.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. URL <https://arxiv.org/abs/2402.03300>.
- Hugo Sousa, Satya Almasian, Ricardo Campos, and Alípio Jorge. Tradutor: Building a variety specific translation model, 2025a. URL <https://arxiv.org/abs/2502.14385>.
- Hugo Sousa, Rúben Almeida, Purificação Silvano, Inês Cantante, Ricardo Campos, and Alípio Jorge. Enhancing portuguese variety identification with cross-domain approaches, 2025b. URL <https://arxiv.org/abs/2502.14394>.
- Fábio Souza, Rodrigo Nogueira, and Roberto Lotufo. Bertimbau: Pretrained bert models for brazilian portuguese. In Ricardo Cerri and Ronaldo C. Prati, editors, *Intelligent Systems*, pages 403–417, Cham, 2020. Springer International Publishing. ISBN 978-3-030-61377-8.
- Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding, 2023. URL <https://arxiv.org/abs/2104.09864>.
- Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. A Bradford Book, Cambridge, MA, USA, 2018. ISBN 0262039249.
- Ye Kyaw Thu, Thazin Myint Oo, and Thepchai Supnithi. RL-nmt: Reinforcement learning fine-tuning for improved neural machine translation of burmese dialects. In *Proceedings of the 5th ACM International Conference on Multimedia in Asia Workshops*, MMAAsia ’23 Workshops, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400703263. doi: 10.1145/3611380.3628564. URL <https://doi.org/10.1145/3611380.3628564>.
- Jörg Tiedemann. *News from OPUS - A Collection of Multilingual Parallel Corpora with Tools and Interfaces*, volume V, pages 237–248. 2009.
- Songjun Tu, Jingbo Sun, Qichao Zhang, Xiangyuan Lan, and Dongbin Zhao. Online preference-based reinforcement learning with self-augmented feedback from large language model, 2024. URL <https://arxiv.org/abs/2412.16878>.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2023. URL <https://arxiv.org/abs/1706.03762>.
- Nuo Xu, Jun Zhao, Can Zu, Sixian Li, Lu Chen, Zhihao Zhang, Rui Zheng, Shihan Dou, Wenjuan Qin, Tao Gui, Qi Zhang, and Xuanjing Huang. Advancing translation preference modeling with rlhf: A step towards cost-effective solution, 2024. URL <https://arxiv.org/abs/2402.11525>.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.

Marcos Zampieri and Binyam Gebre. Automatic identification of language varieties: The case of portuguese. 5, 01 2012a.

Marcos Zampieri and Binyam Gebrekidan Gebre. Automatic identification of language varieties: The case of portuguese. In Jeremy Jancsary, editor, *11th Conference on Natural Language Processing, KONVENS 2012, Empirical Methods in Natural Language Processing, Vienna, Austria, September 19-21, 2012*, volume 5 of *Scientific series of the ÖGAI*, pages 233–237. ÖGAI, Wien, Österreich, 2012b. URL http://www.oegai.at/konvens2012/proceedings/33_zampieri12p/.

Marcos Zampieri, Kai North, Tommi Jauhiainen, Mariano Felice, Neha Kumari, Nishant Nair, and Yash Mahesh Banger. Language variety identification with true labels. In Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue, editors, *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 10100–10109, Torino, Italia, May 2024. ELRA and ICCL. URL <https://aclanthology.org/2024.lrec-main.882/>.

Kaiyan Zhang, Yuxin Zuo, Bingxiang He, Youbang Sun, Runze Liu, Che Jiang, Yuchen Fan, Kai Tian, Guoli Jia, Pengfei Li, Yu Fu, Xingtai Lv, Yuchen Zhang, Sihang Zeng, Shang Qu, Haozhan Li, Shijie Wang, Yuru Wang, Xinwei Long, Fangfu Liu, Xiang Xu, Jiaze Ma, Xuekai Zhu, Ermo Hua, Yihao Liu, Zonglin Li, Huayu Chen, Xiaoye Qu, Yafu Li, Weize Chen, Zhenzhao Yuan, Junqi Gao, Dong Li, Zhiyuan Ma, Ganqu Cui, Zhiyuan Liu, Bqing Qi, Ning Ding, and Bowen Zhou. A survey of reinforcement learning for large reasoning models, 2025. URL <https://arxiv.org/abs/2509.08827>.

Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, Yifan Du, Chen Yang, Yushuo Chen, Zhipeng Chen, Jinhao Jiang, Ruiyang Ren, Yifan Li, Xinyu Tang, Zikang Liu, Peiyu Liu, Jian-Yun Nie, and Ji-Rong Wen. A survey of large language models, 2025. URL <https://arxiv.org/abs/2303.18223>.